

How the 2026 World Cup Prediction Model Works

A complete guide to the mathematical methods behind the joint score probability engine.

Introduction

This model produces a **Probability Mass Function (PMF)** for the final score of every 2026 FIFA World Cup match. For each match the model computes $P(\text{Home goals} = h, \text{Away goals} = a)$ for every possible regulation-time final score, where regulation time is defined as 90 minutes plus stoppage time. Extra time and penalty shootouts are explicitly excluded from all probabilities shown.

Every other number derived from the model — match result probabilities (1X2), totals (over/under), both teams to score (BTTS), team totals, and correct score market prices — is computed by summing the appropriate cells of this single joint distribution. There is no separate model for each market type.

The full grid, including an explicit tail-mass bucket for extreme scores, sums to exactly 1.0. This is enforced at every stage of the pipeline.

Step 1: Rating Every Team — The Composite Prior

Each of the 48 World Cup teams is assigned an **attack lambda** (λ_{att}) and a **defense lambda** (λ_{def}). These represent expected goals scored and conceded against an average opponent at a neutral venue. The 2026 global World Cup average is 1.30 goals per team per match, calibrated from 2018 and 2022 data.

Four rating sources are blended in priority order:

Priority	Source	Weight (market available)	Weight (no market)
1	Market-implied (BDL bookmaker odds)	70%	—
2	Pi rating (penaltyblog)	15%	30%
3	Elo rating (penaltyblog)	10%	45%
4	Massey rating	5%	15%
5	Confederation baseline (floor)	—	10%

Market-implied strength is extracted from bookmaker odds for each team's three group-stage matches using a goal expectancy decomposition: the method reverse-engineers what attack and defense lambdas would produce the observed no-vig 1X2 market probabilities. Up to six bookmaker sources are used per match via the BallDontLie (BDL) API. Not all sources are available for every match.

The Pi and Elo ratings are computed using the penaltyblog library. Pi ratings update continuously with each match result and respond faster to recent form. Elo ratings are more stable over longer periods.

Confederation baselines are applied as a hard floor when rating data is absent. Historical World Cup averages by confederation: CONMEBOL 1.45, UEFA 1.35, CONCACAF 1.20, CAF 1.10, AFC 1.10, OFC 0.90 (attack lambda vs average opponent).

Host advantage: The USA, Canada, and Mexico receive an adjustment of +0.10 to attack lambda and -0.10 to defense lambda as co-hosts. All other matches are treated as neutral venue; no home/away distinction is applied.

Important limitation: the 2018 and 2022 World Cup data together cover 128 matches. This is a limited sample for calibrating team-level ratings, particularly for teams from smaller confederations with few World Cup appearances.

Step 2: The Independent Poisson Model

Given the blended ratings for both teams, the model computes the match-specific expected goals for each side:

$$\lambda_{\text{home}} = \text{home_attack} \times \text{away_defense} \times \text{global_scale_factor} \quad \lambda_{\text{away}} = \text{away_attack} \times \text{home_defense} \times \text{global_scale_factor}$$

The number of goals each team scores is modeled as an independent **Poisson distribution**. The Poisson distribution describes the probability of a given count of events occurring in a fixed time interval when those events occur at a constant average rate and independently of one another. For goals in a football match this is an approximation — goals are not entirely independent (a goal changes the incentives for both teams) — but it is the standard model in the football forecasting literature and performs well on held-out data.

The joint probability of a specific final score is:

$$P(\text{Home} = h, \text{Away} = a) = [e^{(-\lambda_{\text{home}})} \times \lambda_{\text{home}}^h / h!] \times [e^{(-\lambda_{\text{away}})} \times \lambda_{\text{away}}^a / a!]$$

This is computed for all integer pairs (h, a) from (0,0) up to the grid maximum (typically 8 or 9 goals per team). The remaining probability mass — for scores beyond the grid — is captured in an explicit tail-mass value shown alongside every prediction. The independence assumption is partially corrected in the market reconciliation step that follows.

Step 3: Market Reconciliation

Market odds contain information that the pure Poisson model does not have access to: late team news, tactical changes, injury reports not yet public, and the aggregate view of professional traders. The reconciliation step incorporates this information.

For each match, no-vig probabilities are extracted from all available bookmaker markets: match result (1X2), correct score lines, total goals over/under (0.5, 1.5, 2.5, 3.5), both teams to score, draw no bet, double chance, and team-level totals. The bookmaker margin is removed from all odds before use.

A constrained optimization (SLSQP algorithm) then adjusts the initial Poisson PMF grid to satisfy these market constraints while minimizing divergence from the original distribution. The optimization operates on an 8×8 grid of joint probabilities subject to the constraint that all values remain non-negative and sum to 1.0.

The result is the **market-reconciled PMF**: a distribution that respects both the model's fundamental strength estimates and the market's current pricing. When market odds are unavailable for a match (as may occur in

the early tournament stages before lines are posted), the pure Poisson model is used directly.

Step 4: Calibration

Calibration refers to the alignment between stated probabilities and observed frequencies. A calibrated model is one where outcomes it assigns 30% probability occur approximately 30% of the time over many predictions.

The model uses **temperature scaling**: a single parameter T is fitted by minimizing exact-score log loss on out-of-sample predictions from the 2018 and 2022 World Cups. The adjusted probability is:

$$p_{\text{calibrated}}[h,a] \propto p_{\text{raw}}[h,a] ^ (1/T)$$

$T > 1$ indicates the model is overconfident and flattens the distribution. $T < 1$ indicates underconfidence and sharpens it. $T = 1$ leaves the distribution unchanged.

Calibration is evaluated using multiple metrics computed on out-of-sample predictions only (never on the training data):

Exact-score log loss: Primary metric. Measures how surprised the model was by actual final scores. Lower is better.

Ranked Probability Score (RPS): Standard metric for ordered categorical outcomes. Applied to the 1X2 market.

Brier score: Mean squared error on binary market probabilities (BTTS, over/under).

Expected Calibration Error (ECE): Direct measure of how often stated probabilities match observed frequencies across probability bins.

Ignorance score: Log score for probabilistic forecasts.

Note: with only 128 matches across two tournaments available for calibration, these metrics carry meaningful uncertainty. The calibration temperature is a coarse correction, not a guarantee of statistical accuracy.

Step 5: Edge Screening

Once a calibrated PMF is available, the model compares its implied probabilities against the market's no-vig probabilities for every available bet. The edge on any outcome is defined as:

$$\text{Edge} = (\text{Model probability} - \text{Market implied probability}) / \text{Market implied probability}$$

A market is flagged as a potential value opportunity only when all three of the following conditions hold simultaneously:

- Edge $\geq 4\%$ (the minimum threshold is 0.04, hardcoded in the engine)
- The lower bound of the model's 90% confidence interval still exceeds the market implied probability. The CI is computed by perturbing λ by $\pm 12\%$ (a conservative assumed uncertainty on the lambda estimates) and taking the resulting range of PMF-derived probabilities.
- Market implied probability $> 2\%$ (outcomes with under 2% market-implied probability are excluded due to thin liquidity and high variance on the edge estimate)

Bets passing all three filters are displayed as value opportunities. The engine computes a **half-Kelly stake** for each:

```
f* = edge / (decimal_odds - 1) [Half-Kelly = f* / 2] Capped at 5% of bankroll
```

Half-Kelly is used rather than full Kelly because the 12% lambda uncertainty assumption is approximate. Overestimating the edge with full Kelly results in overbetting. The 5% cap is an additional hard limit regardless of computed edge.

The Daily Pipeline

An automated process runs at 4:30 AM UTC each day via a server-side timer. GitHub Actions runs as a fallback at 4:00 AM UTC and skips the pipeline if the server is confirmed healthy. The pipeline stages are:

Fetch: Pull the latest match records, bookmaker odds (up to six sources), team statistics, expected goals, shot maps, match events, momentum data, and group standings from the BallDontLie API.

Build: Rebuild versioned Parquet tables from raw API snapshots, catching schema changes.

Predict: Run the composite prior, Poisson model, market reconciliation, calibration, and edge screening for every upcoming match.

Publish: Write one JSON file per matchday containing the full PMF grid, all derived markets, the edge report, and calibration metadata.

Validate: Verify internal consistency: PMF sums to $1.0 \pm 1e-6$, no probabilities outside $[0,1]$, no missing required fields.

Deploy: Upload prediction data and HTML pages to the production server via FTP.

Report: Generate a matchday summary including group standings, advancement probabilities, and top value opportunities.

During match hours (approximately 14:00–04:00 UTC), a separate live process runs every minute to poll BallDontLie for live match states and update the in-play PMF. When the production server's WebSocket endpoint is active, updates are pushed directly to connected browsers.

Limitations and Scope

- All probabilities represent regulation time only (90 minutes plus stoppage time). Extra time and penalty shootouts are not modeled.
- The Poisson independence assumption is an approximation. Teams adjust tactically after goals, creating correlation between home and away goal counts. Market reconciliation partially corrects for this but does not eliminate it.
- The calibration dataset covers 128 World Cup matches (2018 and 2022 combined). This is a small sample and calibration metrics carry meaningful uncertainty.
- The $\pm 12\%$ lambda uncertainty used for confidence intervals is a fixed assumption, not an empirically derived estimate for each match.
- Odds used for edge calculations are sourced at the time the daily prediction is generated. Odds move before kickoff. Always verify current odds before acting on any edge signal.
- Edge estimates are outputs of a probabilistic model. They are not guaranteed profit signals.

All probabilities represent regulation time (90 minutes + stoppage time) only. Extra time and penalties are excluded. This article is for informational and educational purposes. Please gamble responsibly.